SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50

Code of Conduct:

I P.Nikitha/192324226(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

P. Nikitha

Signature of Student:

Annexure A
CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

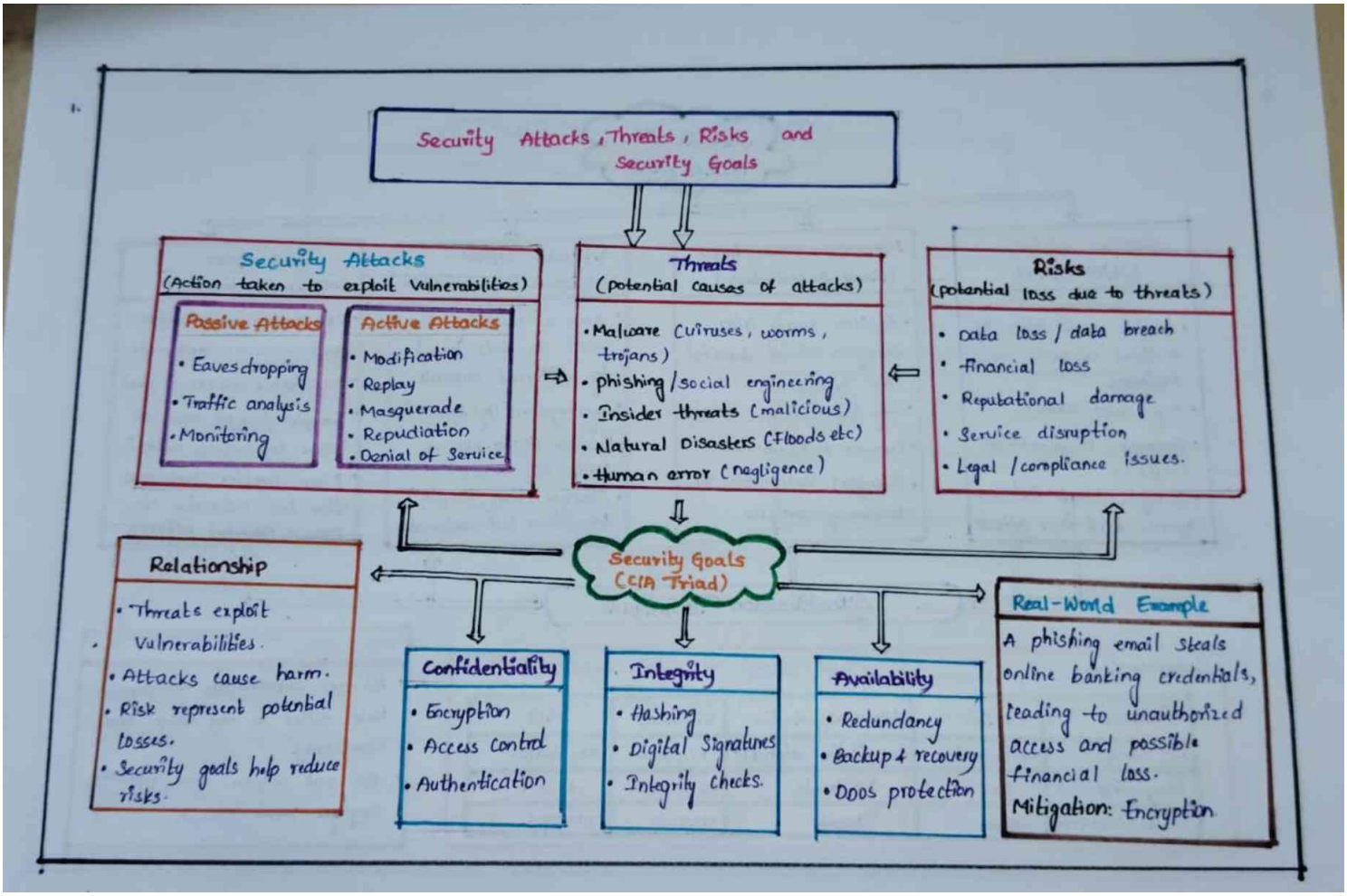
a. Show how each attack leads to specific threats and risks
b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why. 3. Construct a concept map for Classical Encryption Schemes: Substitution
Techniques and Transposition Techniques. Perform the following tasks:
a. Show differences and similarities

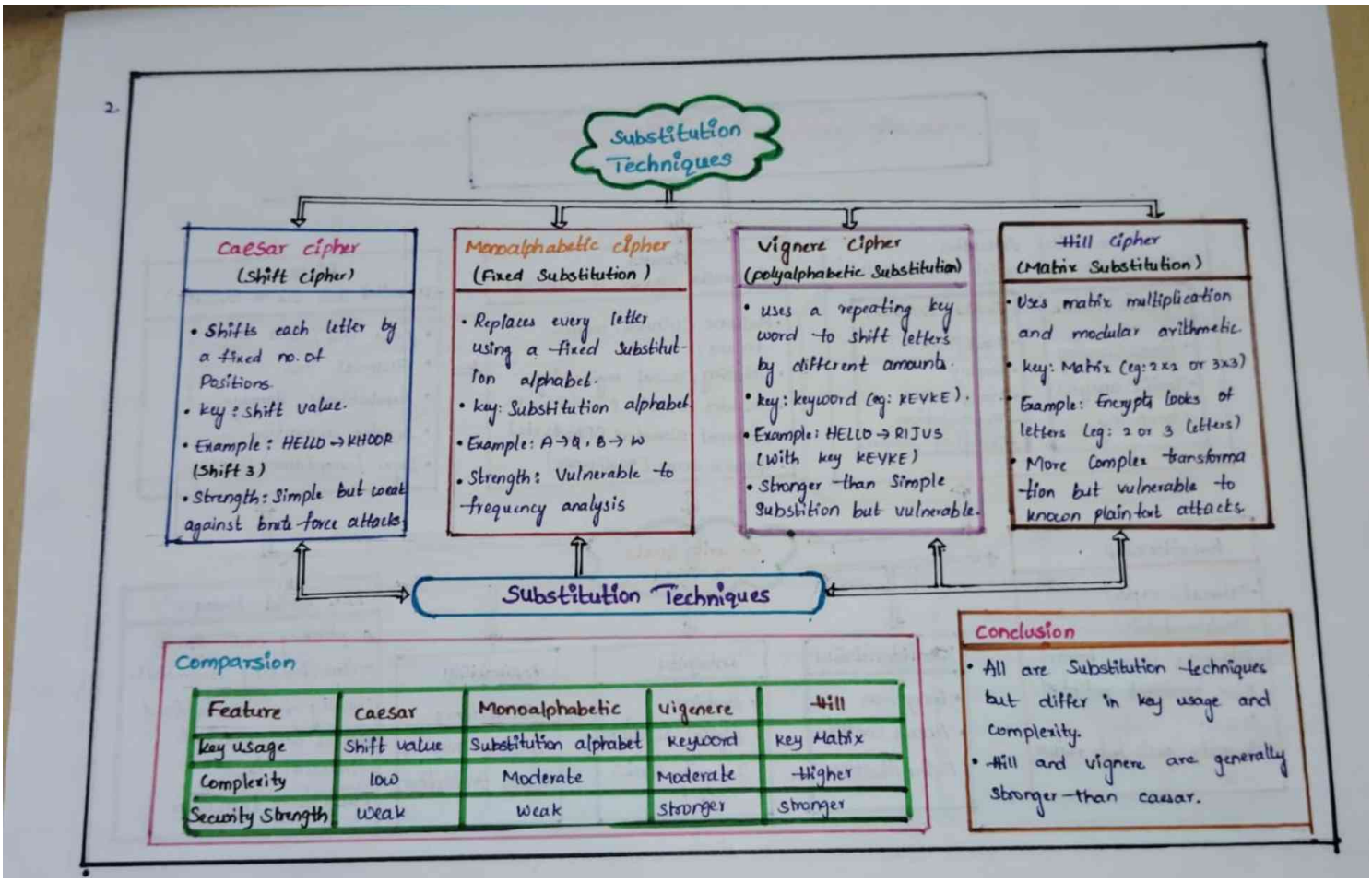
i.Map how each affects the Data structure and Security level
ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

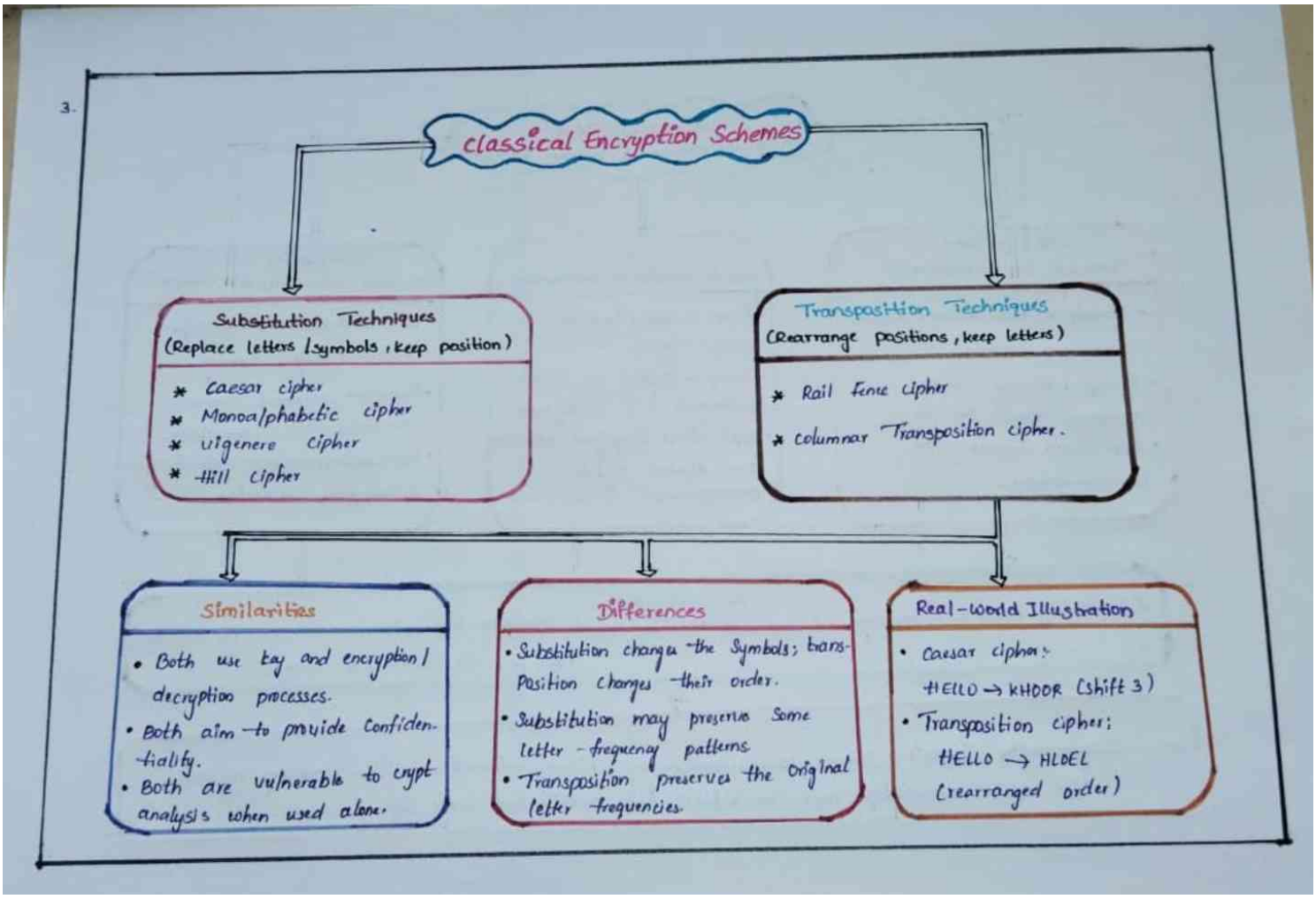
a. Show relationships between these concepts
b. Analyze how each contributes to secure communication
c. Include one real-world application for each

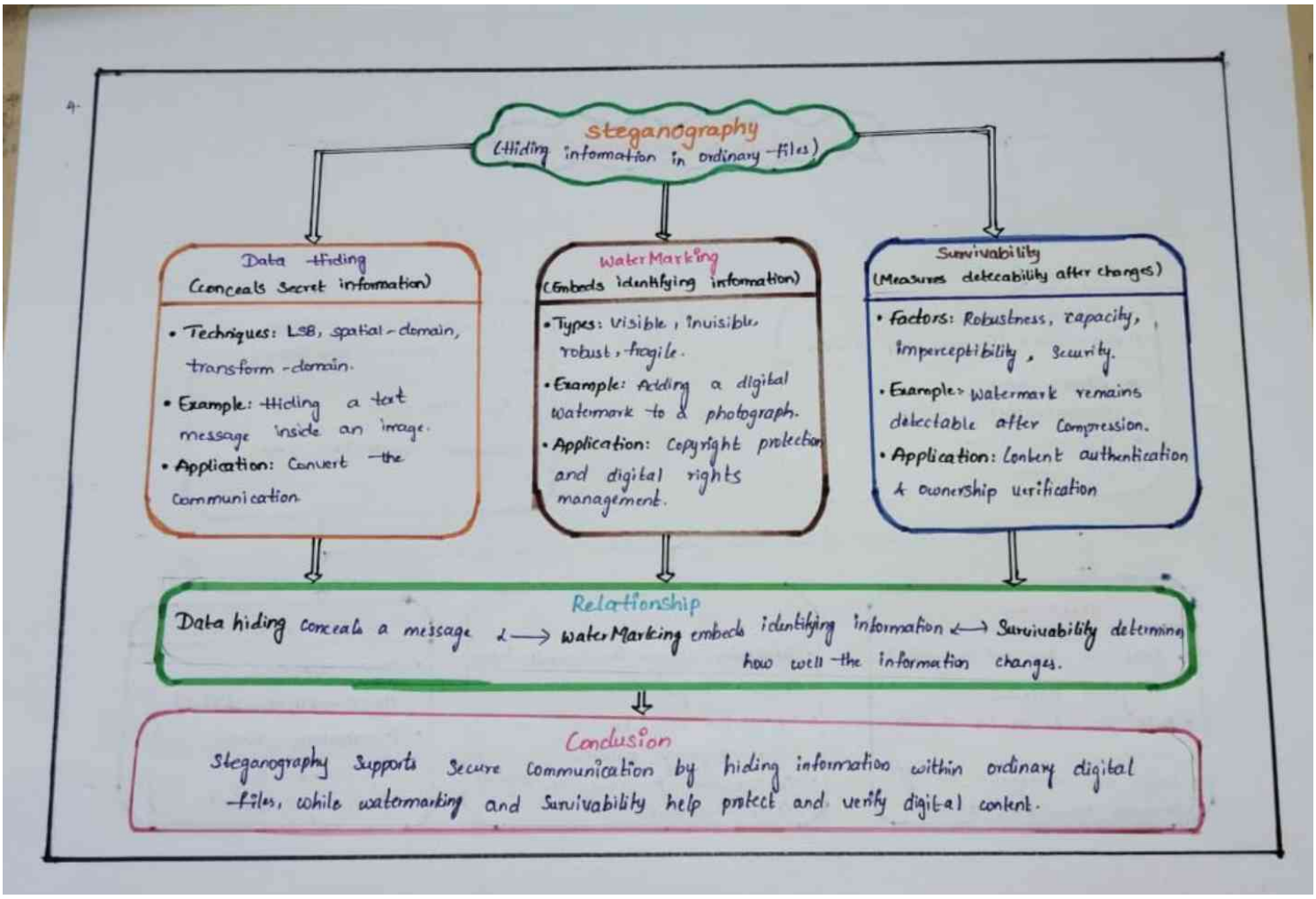

Security Attacks, Threats, Risks and Security Goals
Security Attacks
(Action taken to exploit Vulnerabilities)
Passive Attacks
• Eavesdropping
• Traffic analysis
• Monitoring
Active Attacks
• Modification
• Replay
• Masquerade
• Repudiation
• Denial of Service
Threats
(potential causes of attacks)
• Malware (viruses, worms, trojans)
• phishing / social engineering
• Insider threats (malicious)
• Natural Disasters (Floods etc)
• Human error (negligence)
Risks
(potential loss due to threats)
• Data loss / data breach
• Financial loss
• Reputational damage
• Service disruption
• Legal / compliance issues.
Security Goals
(CIA Triad)
Relationship
• Threats exploit Vulnerabilities.
• Attacks cause harm.
• Risk represent potential losses.
• Security goals help reduce risks.
Confidentiality
• Encryption
• Access control
• Authentication
Integrity
• Hashing
• Digital Signatures
• Integrity checks.
Availability
• Redundancy
• Backup & recovery
• DDOS protection
Real-World Example
A phishing email steals online banking credentials, leading to unauthorized access and possible financial loss.
Mitigation: Encryption

2.


Substitution Techniques
Caesar cipher (Shift cipher)
• Shifts each letter by a fixed no. of positions.
• key : shift value.
• Example : HELLO → KHOOR (Shift 3)
• Strength : Simple but weak against brute-force attacks.
Monoalphabetic cipher (Fixed Substitution)
• Replaces every letter using a fixed substitution alphabet.
• key: Substitution alphabet
• Example: A → Q, B → W
• Strength : Vulnerable to frequency analysis
Vignere Cipher (polyalphabetic Substitution)
• uses a repeating key word to shift letters by different amounts.
• key : keyword (eg: KEYKE).
• Example: HELLO → RIJUS (with key KEYKE)
• Stronger than simple substition but vulnerable.
Hill cipher (Matrix Substitution)
• Uses matrix multiplication and modular arithmetic.
• key: Matrix (eg: 2x2 or 3x3)
• Example: Encrypts looks of letters (eg: 2 or 3 letters)
• More complex transformation but vulnerable to known plaintext attacks.
Substitution Techniques
Comparsion
Feature | caesar | Monoalphabetic | vigenere | Hill
key usage | Shift value | Substitution alphabet | Keyword | key Matrix
Complexity | low | Moderate | Moderate | Higher
Security Strength | weak | weak | Stronger | Stronger
Conclusion
• All are Substitution techniques but differ in key usage and complexity.
• Hill and vignere are generally stronger than caesar.

3.


Classical Encryption Schemes
Substitution Techniques
(Replace letters / symbols, keep position)
* Caesar cipher
* Monoalphabetic cipher
* Vigenere cipher
* Hill cipher
Transposition Techniques
(Rearrange positions, keep letters)
* Rail Fence cipher
* columnar Transposition cipher.
Similarities
• Both use key and encryption / decryption processes.
• Both aim to provide confidentiality.
• Both are vulnerable to crypt analysis when used alone.
Differences
• Substitution changes the Symbols; transposition changes their order.
• Substitution may preserve some letter-frequency patterns.
• Transposition preserves the original letter frequencies.
Real-world Illustration
• Caesar cipher:
HELLO → KHOOR (shift 3)
• Transposition cipher:
HELLO → HLOEL
(rearranged order)


4.
Steganography
(Hiding information in ordinary files)
Data hiding
(Conceals secret information)
• Techniques: LSB, spatial-domain, transform-domain.
• Example: Hiding a text message inside an image.
• Application: Convert the communication.
Watermarking
(Embeds identifying information)
• Types: visible, invisible, robust, fragile.
• Example: Adding a digital watermark to a photograph.
• Application: Copyright protection and digital rights management.
Survivability
(Measures detectability after changes)
• Factors: Robustness, capacity, imperceptibility, security.
• Example: Watermark remains detectable after compression.
• Application: Content authentication & ownership verification
Relationship
Data hiding conceals a message ⟷ Watermarking embeds identifying information ⟷ Survivability determining how well the information changes.
Conclusion
Steganography supports secure communication by hiding information within ordinary digital files, while watermarking and survivability help protect and verify digital content.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding or misinterpretation of the problem
Experimental Design	20	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Fails to apply appropriate concepts
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	25	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	interpretation Basic analysis with limited	Incorrect or incomplete solution
Documentation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation